[GCT555 3DID]

ProxiWall: Distance-Driven Semantic Zoom for HMD-Free Virtual Museum Exploration on Wall Display

20265015 Heejeong Ko
20215506 JungJin Park
20263574 Seungjae Lee

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Problem & Need : UX Scenario

- General museum visitors need an accessible, shared, and intuitive way to explore virtual museum spaces without wearing HMDs or using controllers.
- **Problem:** A fixed wall display has only one viewpoint — yet far-viewers and near-viewers each want a different one. Without an HMD, today's wall systems cannot serve both.

FAR · Overview — visitors scan the room, taking in many works at a glance.

NEAR · Detail — visitors stp, focus, and examine a single work.

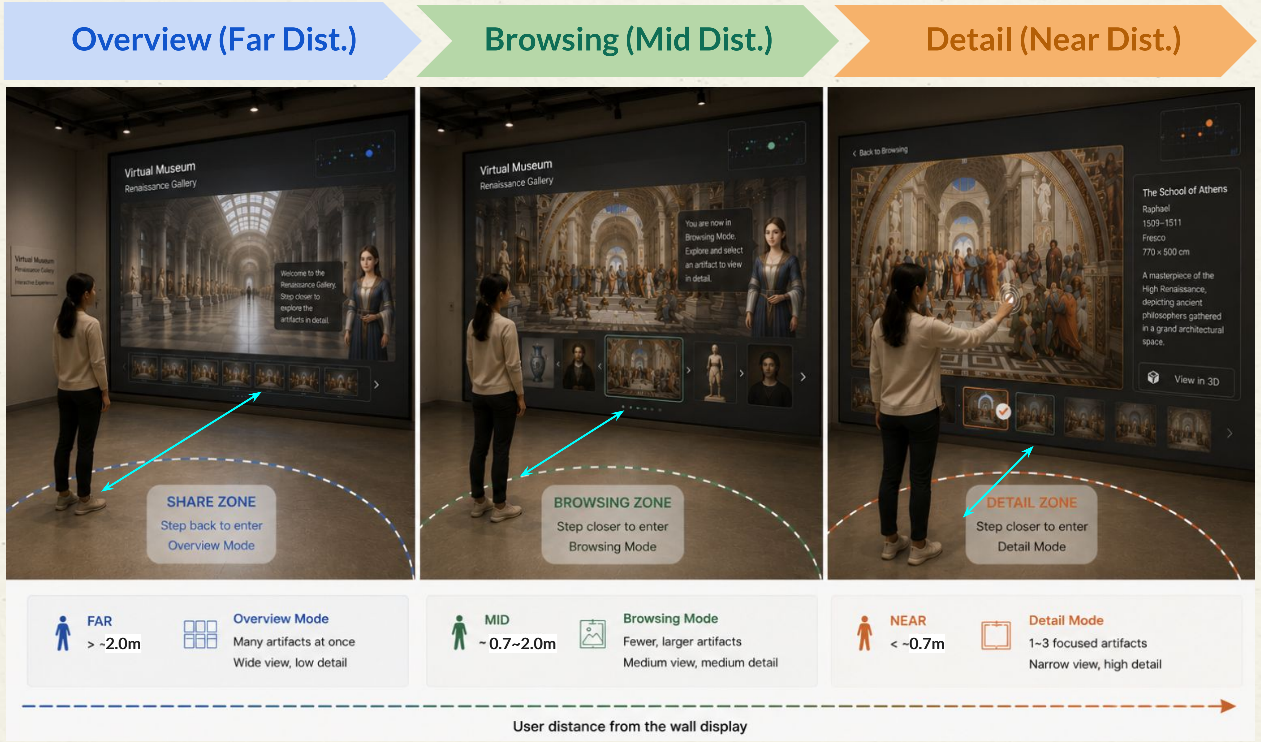
▸ **[The core idea]:** Museum visitors should be able to explore virtual galleries the way they explore real ones — by **walking and looking** — without strapping on a headset.

Approach

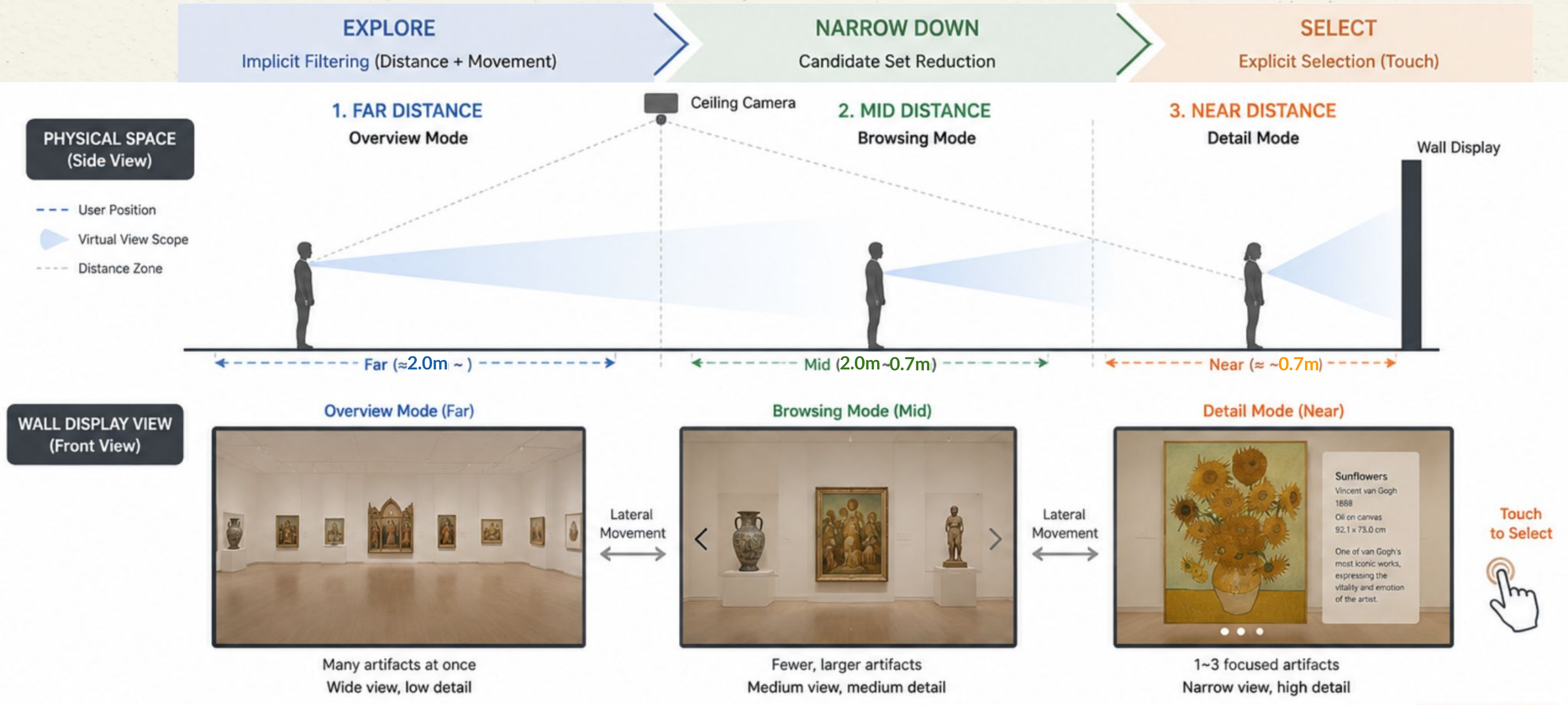
- **Environment setting**
    a. Input Multiple RGB-D Camera (Front view + Top view)
    b. Output 3D Rendered Video — LED Wall (w/o HMD)
- **Interaction feature**
    a. **Distance-based Semantic Zoom** : User distance from the wall automatically changes the view mode
        i. Distance controls Overview → Browsing → Detail transition
    b. **Body-based Gallery Navigation:** Lateral movement controls horizontal exploration
    c. **Pseudo touch-based selection** : Touch confirms the final artifact selection

Distance-based Exploration

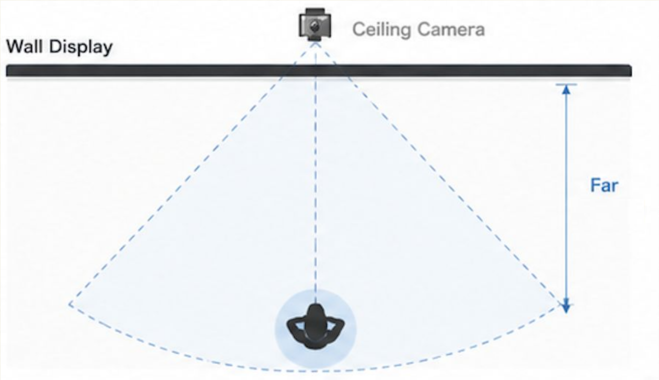
User distance from the wall automatically changes the view mode.



Exploration Mode (Overview, Browsing, Detail)



[Far] Overview mode



Overview

Peripheral + UFOV scanning · n bounded by Miller 7±2 (loose) · each obj at ~5–7° angular size for floor recognizability Greenberg et al.'s Proxemic Interactions *(2011)* ; Miller 1956 ; Vogel & Balakrishnan 2004]

1

Overview

~2.0 m
FoV 53°
n=8–10

2

Browsing

0.7m~2.0 m
FoV 53~110°
n=4–6

3

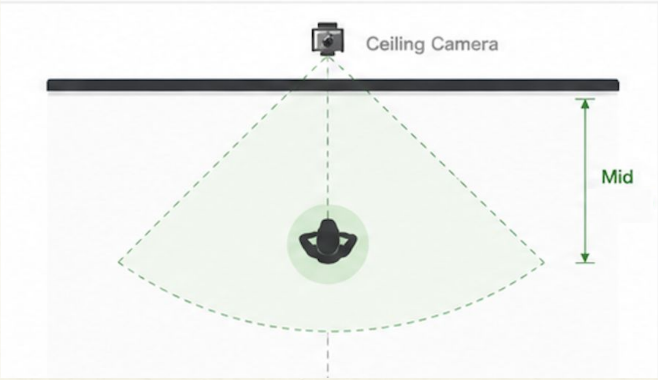
Detail

≤0.7 m + wall touch
FoV 110°
n=1–3

Wall Display

Distance (m)

[Mid] Browsing mode



Browsing

Dynamic foveal-parafoveal compare zone · n reduced per Hick-Hyman $\log_2(n+1)$ · fits 1–2 saccade cycle [Miller 1956 ; Hick 1952 ; Hyman 1953]

1

Overview

~2.0 m
FoV 53°
n=8–10

2

Browsing

0.7m~2.0 m
FoV 53~110°
n=4–6

3

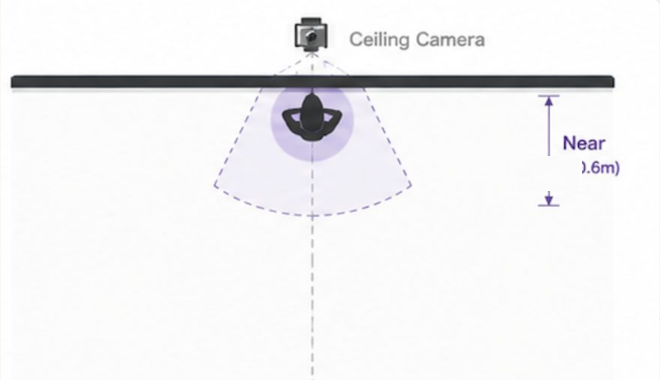
Detail

≤0.7 m + wall touch
FoV 110°
n=1–3

Wall Display

Distance (m)

[Near] Detail mode



Detail

1 obj subtends > 50° → foveal / ventral identification · Hick-Hyman RT ≈ 0 at n = 1–3 · Fitts target W maximized for explicit commit [Goodale & Milner 1992 ; Fitts 1954 ; Dourish 2001]

1
Overview
~2.0 m
FoV 53°
n=8–10

2
Browsing
0.7m~2.0 m
FoV 53~110°
n=4–6

3
Detail
≤0.7 m + wall touch
FoV 110°
n=1–3

Wall Display

Distance (m)

Comparisons

	ProxiWall	CollabVisAdapt (2026/VR)	BlanchTouch (2026/VR)	Co-Living between Reality & Virtuality (2026/CHI)
Main Focus	Collaborative 3D exploration	Distance-adaptive MR visualization	Vision-based wall touch	HMD-free virtual–physical integration
Interaction	Gesture + pseudo touch + sound	Distance-based adaptation	Fingertip touch detection	Natural body/touch interaction
Core Idea	Shared ↔ Private switching	Visualization switching	Surface touch sensing	Reality–virtuality coexistence
Key Difference	Multimodal collaborative workflow	Rendering adaptation only	Touch input only	Conceptual framework
Display	Dual LED wall	HMD AR/VR	HMD + RGB camera	Spatial / wall displays

Research Question

Main RQ:

Are *distance-based body input* and *direct wall touch* meaningful interaction modalities for **the *Overview* → *Browsing* → *Selection* funnel** of a Wall Display virtual museum?

- OVERVIEW Stage (~2.0 m, FoV ≈ 37°, n=8-10)
- BROWSING Stage (0.7m~2.0 m, FoV ≈ 67°, n=4-6)
- SELECTION Stage (≤0.7 m + wall touch, n=1-3)

Hypotheses

H1 The proposed interaction will result in significantly shorter **Task Completion Time** compared to the baseline *(Controller-based Manipulation)*.

H2 The proposed interaction will result in significantly higher **Selection Accuracy** compared to the baseline.

H3 The proposed interaction will result in significantly lower **Selection Error Rate** compared to the baseline.

H4 The proposed interaction will improve **Exploration Efficiency**, as reflected by fewer non-target steps before reaching the correct target compared to the baseline.

H5 The proposed interaction will result in lower perceived cognitive workload, as measured by **NASA-TLX** and higher **Spatial Presence** as measured by MEC-SPEQ compared to the baseline.

Evaluation

Objective Measures

Metric	Measurement
Task Completion Time	Time from Task Start to Target Artifact Selection (seconds)
Exploration Efficiency	Number of non-target artifacts explored or interaction steps before correct target selection.
Selection Accuracy	Percentage of trials in which the participant correctly selected the target artifact.
Selection Error Rate	Percentage of trials involving incorrect object selection or failure to select the target artifact.

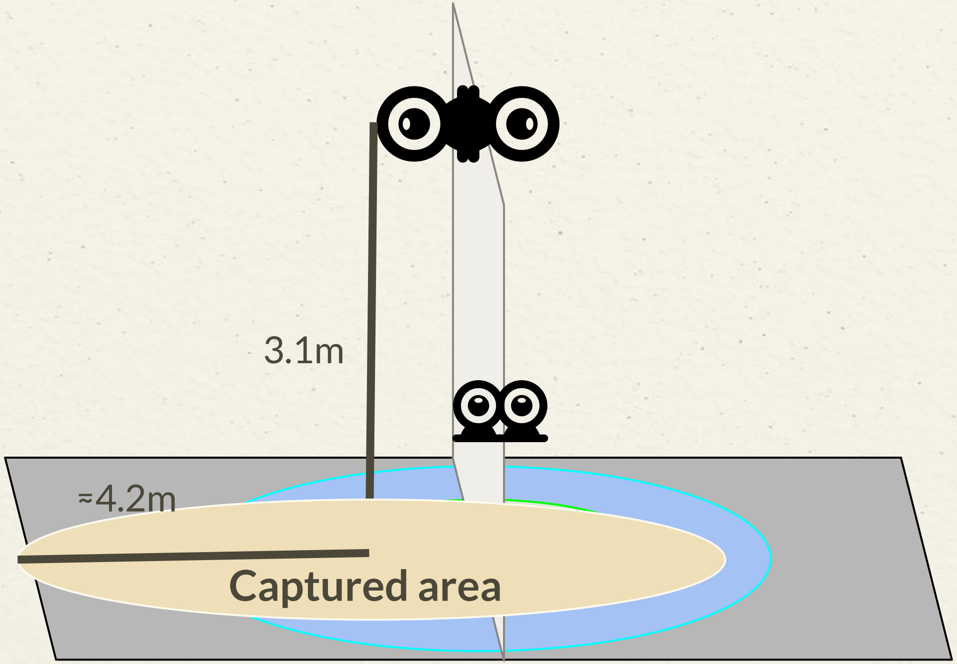
Subjective Measures

Metric	Measurement
NASA-TLX	Workload, mental demand, and physical demand
Presence (MEC-SPQ, 8 items)	Spatial situation model, self-location, and possible actions in the virtual museum

Qualitative Evaluation

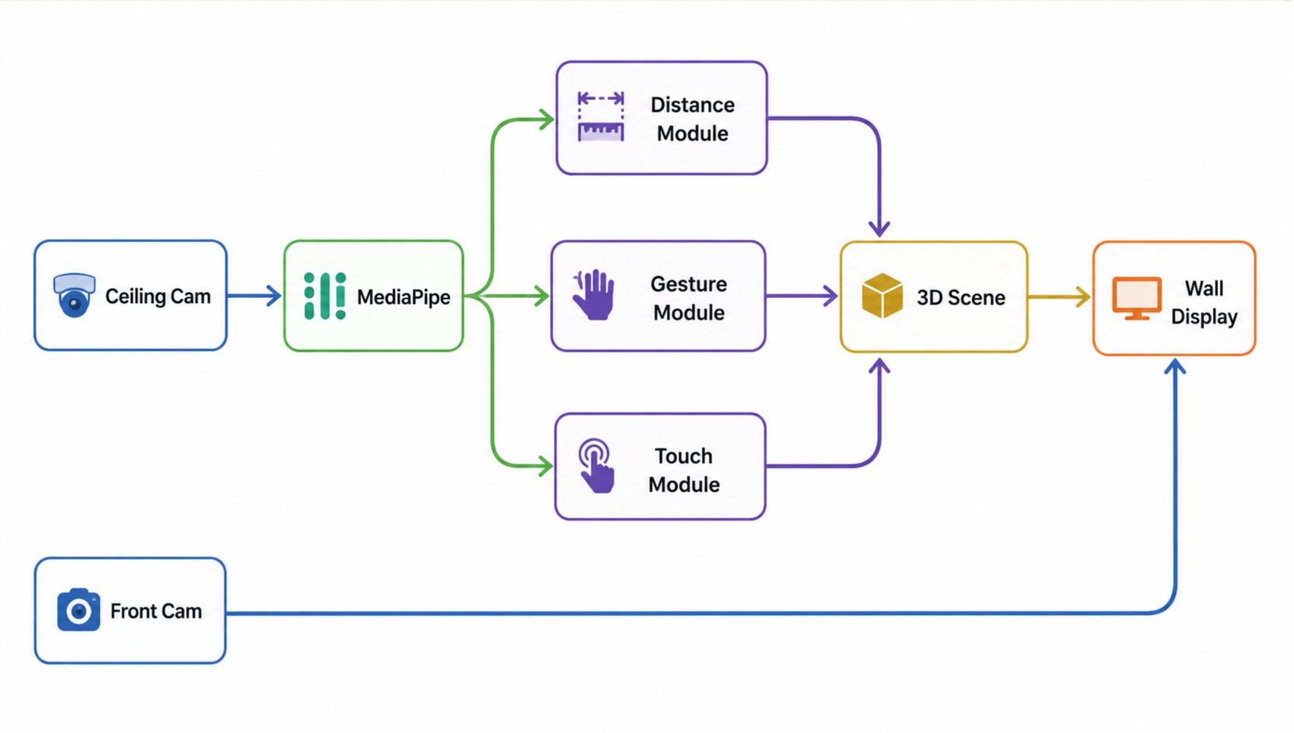
Semi-Structured Interview	Naturalness, immersion, discomfort, and **preference**

System Setup

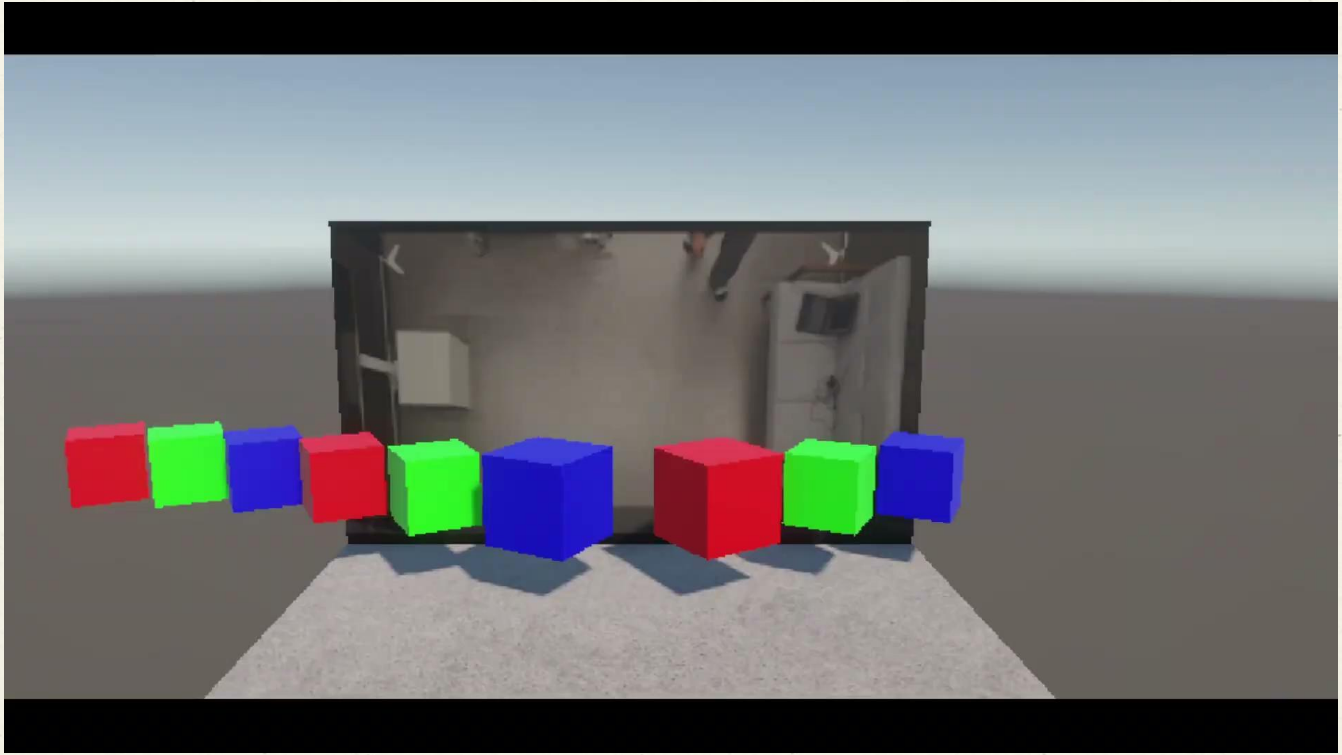


= Overview Mode
= Browsing Mode
= Detail Mode
= LED Wall
14mm crop body camera

Implementation



Demo.



Timeline

Description	Week										Status
	7	8	9	10	11	12	13	14	15	16	
Interaction Concept											Completed
Tracking Setup											Completed
Unity Prototype											Completed
Parameter Design for Interaction											In Progress
Distance-Based Semantic Zoom											In Progress
FoV-Based Object Mapping											Not started
User Study											Not started
Analysis											Not started

Heejeong Seungjae Jungjin

Thanks!

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